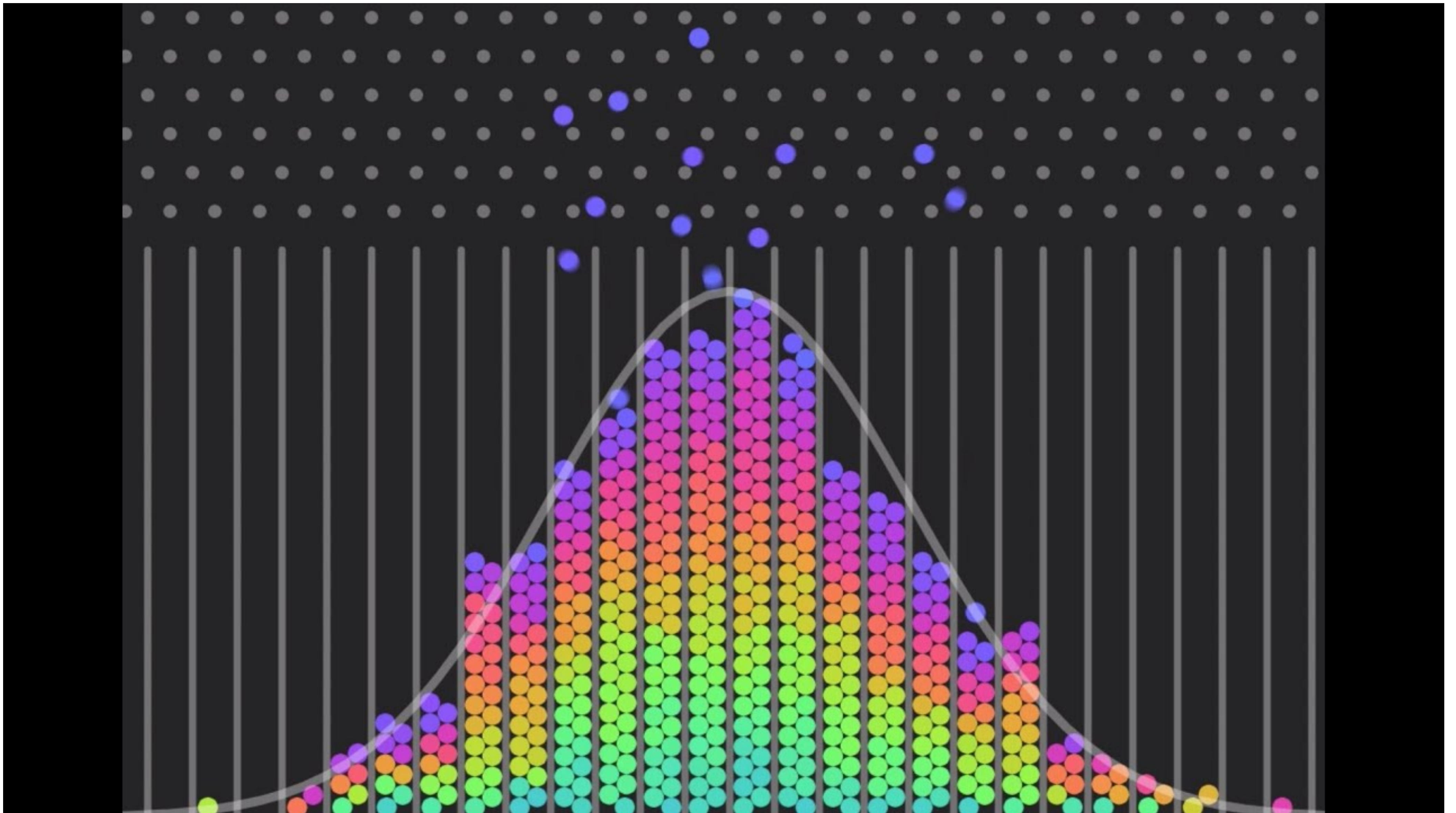


PC5253 Complex System Analysis and Modelling

- The supposedly wide-spread normal distribution due to central limit theorem

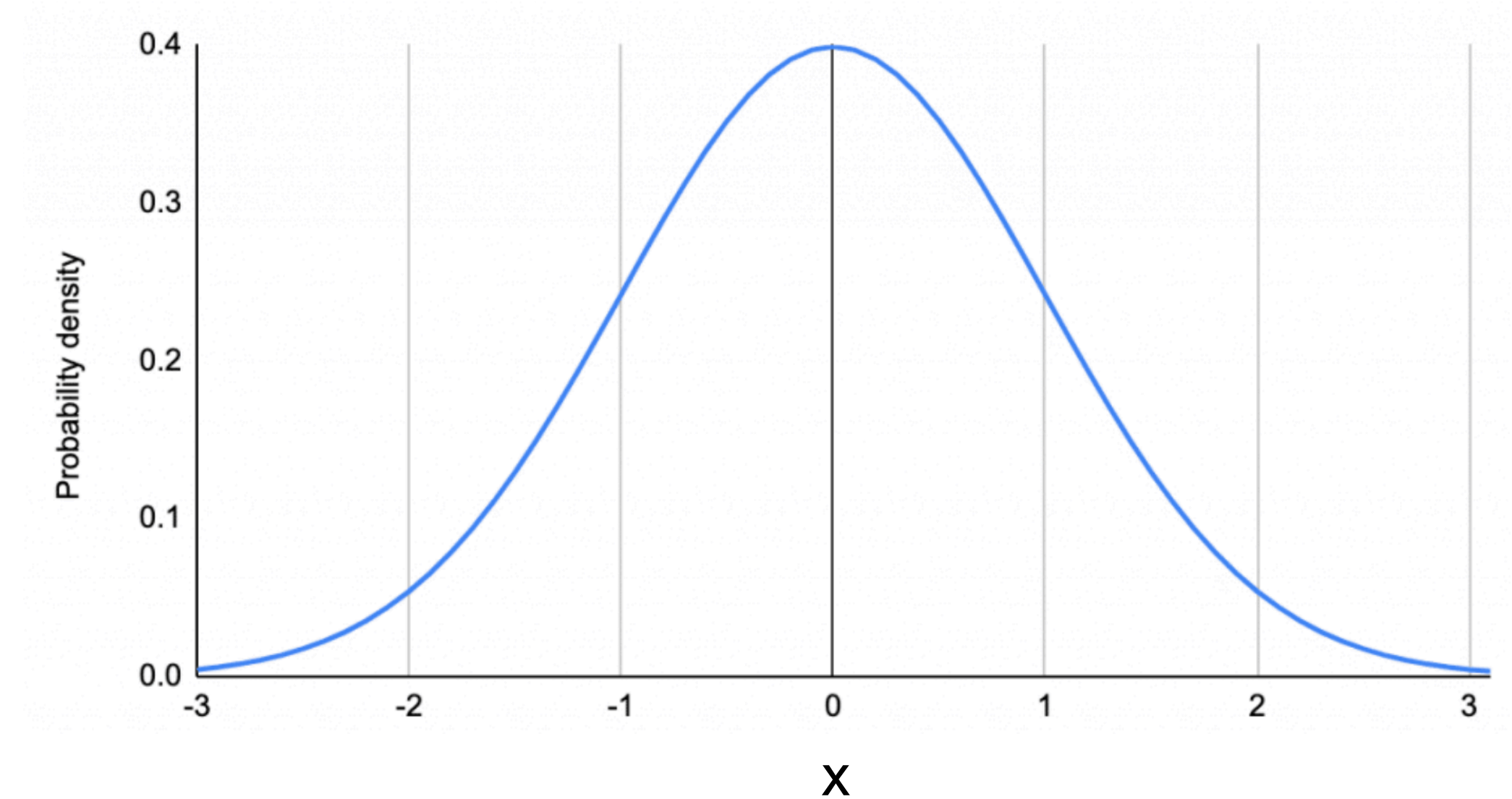
* Reference: “An introduction to Econophysics” By Rosario N. Mantegna and H. Eugene Stanley, Chapter 3.



<https://www.youtube.com/watch?v=DxunFkBDdTA>

What is normal distribution and central limits theorem?

Probability density function (**PDF**):

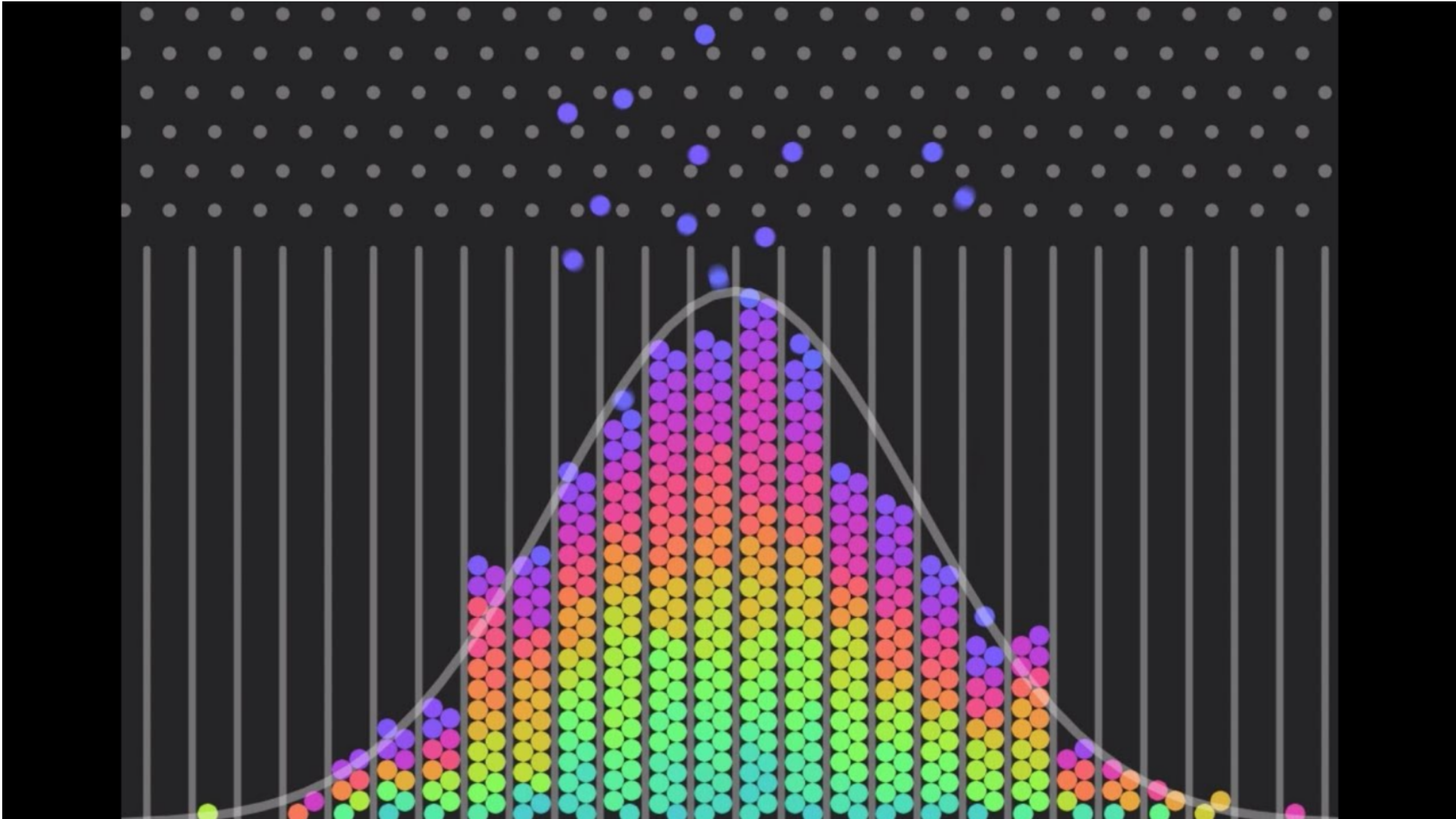
$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$


Central limit theorem: When **many INDEPENDENT RANDOM** values add up, their sum follows ***normal distribution***
Condition: the random variables have identical and finite mean and variance.

Mathematical description:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \longrightarrow p(\sqrt{n}\bar{x}) = \frac{1}{\sqrt{2\pi}} e^{-(\sqrt{n}\bar{x})^2/2}$$

Many
Identical
Independent
Random
Sum
Probability

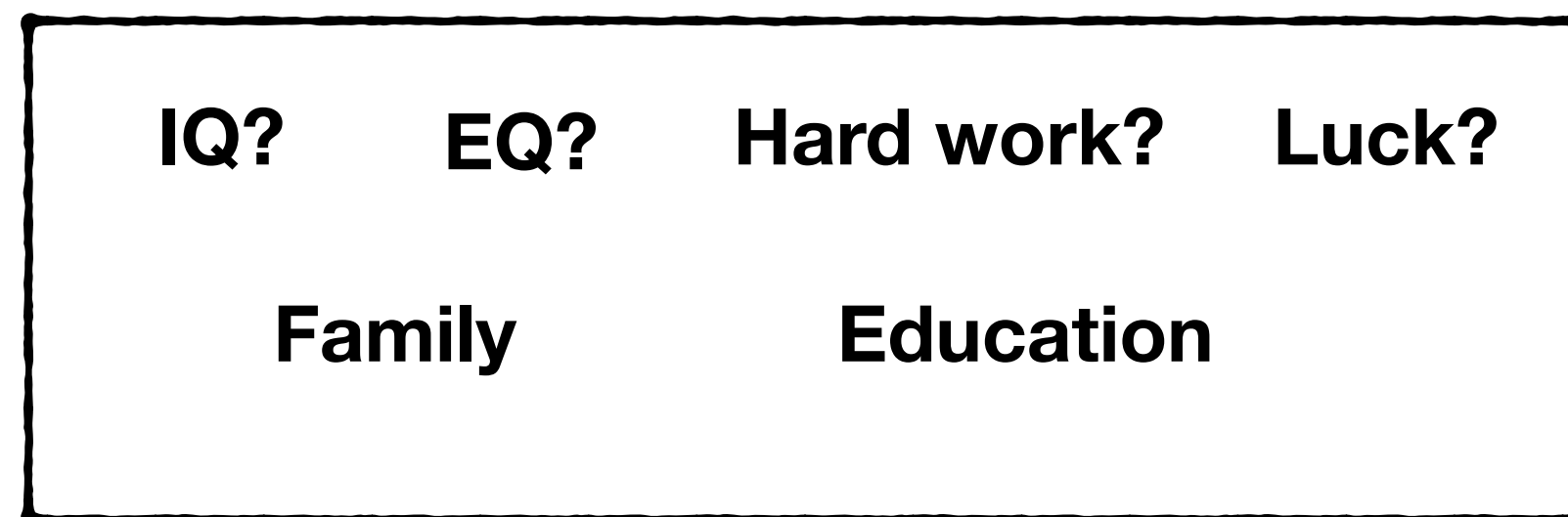


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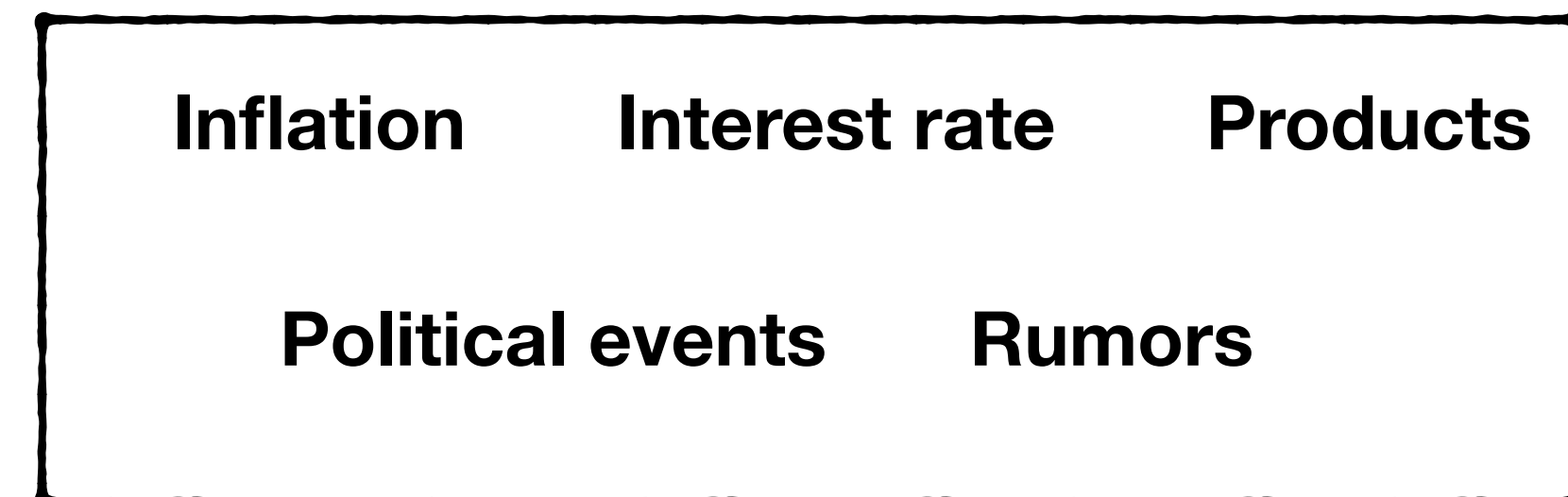
Why is it important?

*Many real world quantities can be assumed (?) as the outcome of many different factors (**complex?**) adding up - so are expected to converge to normal distribution. Hence, when we look at the distribution of these quantities, **normal distribution** should be a good first guess.*

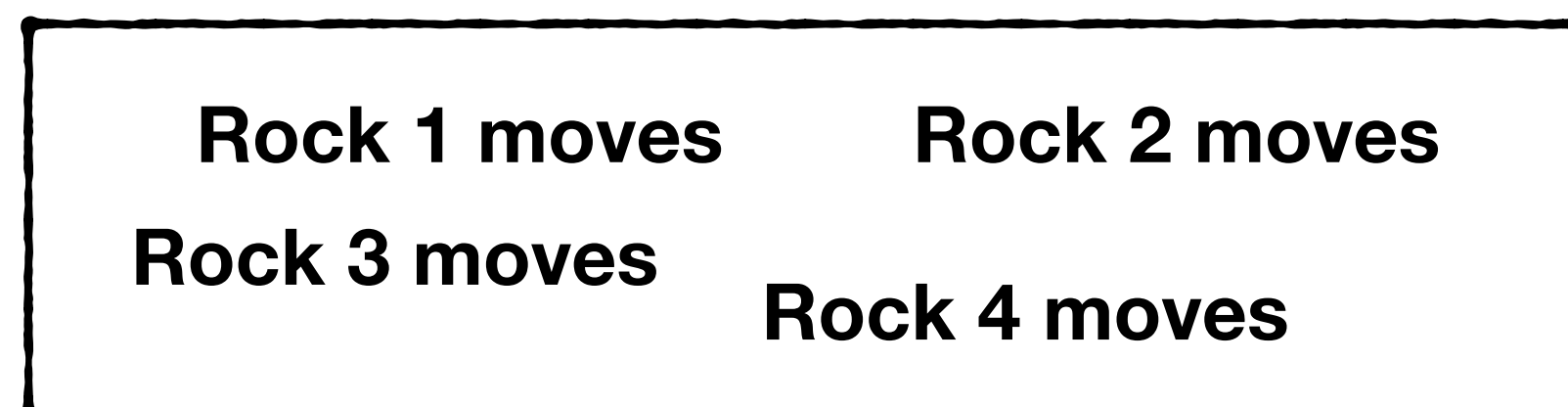
Personal wealth



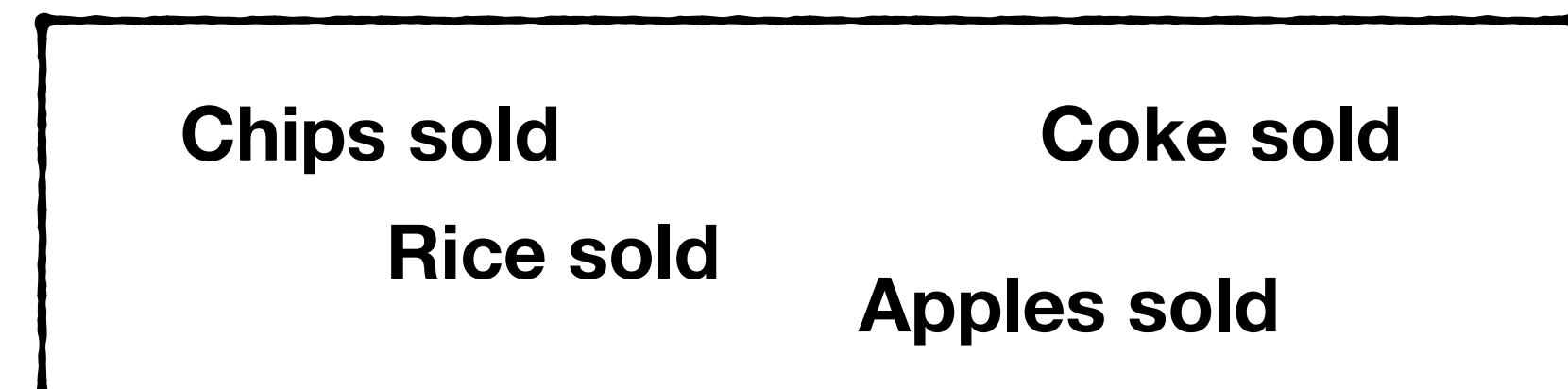
Stock price



Earthquakes



Super market total sales



Central limit theorem

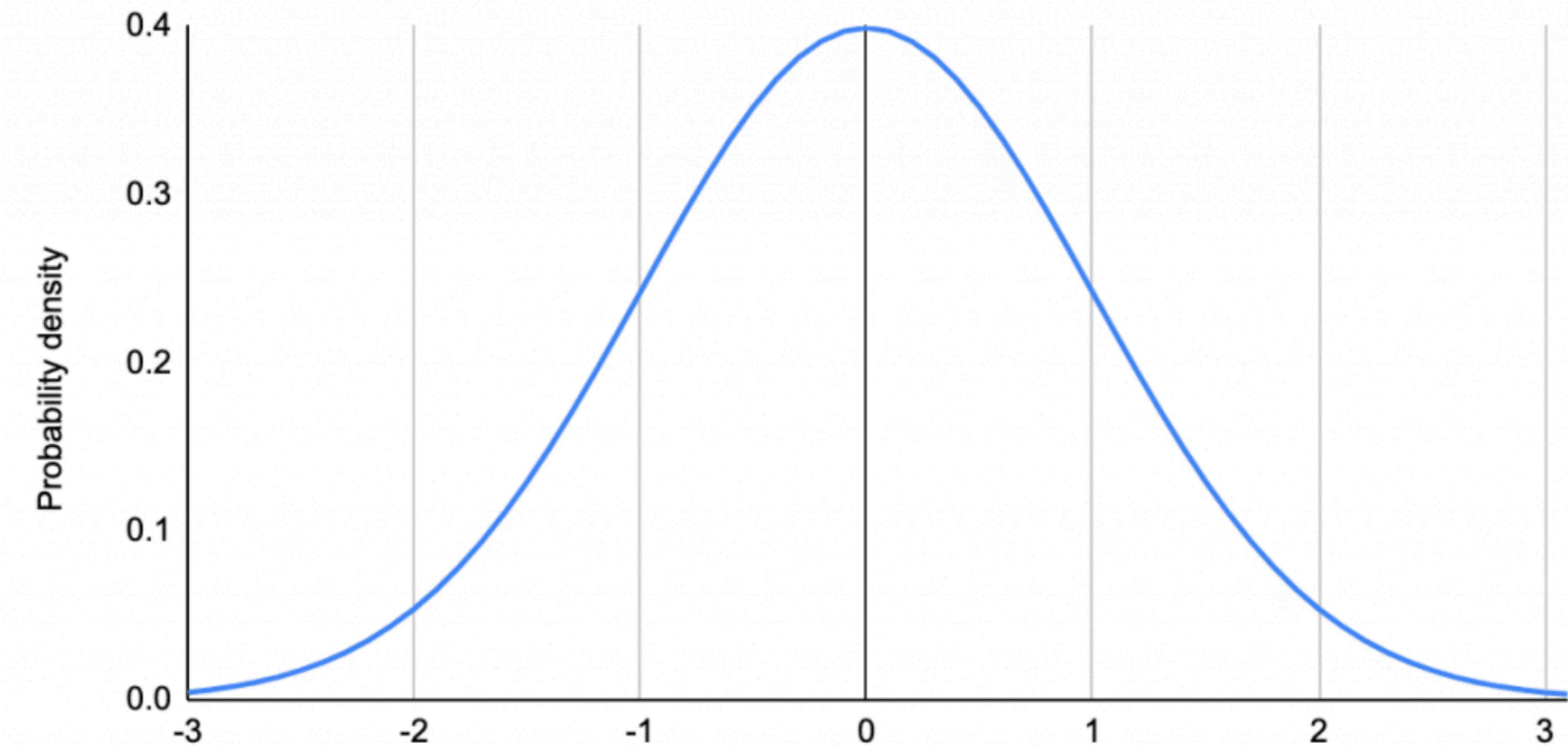
- Basic concepts of distributions

Probability density function (**PDF**):
(Standard normal distribution)

$$p(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}$$

Total probability must be 1

$$\int_{-\infty}^{+\infty} p(x) dx = 1$$



Mean (average) value of distribution $\mu_x = E[x] = \int_{-\infty}^{+\infty} xp(x) dx = 0$

Variance of distribution

$$\sigma_x^2 = E[x^2] - (E[x])^2 = \int_{-\infty}^{+\infty} x^2 p(x) dx - \mu_x^2 = 1$$

Central limit theorem

- Exact statement

Assumptions: i.i.d. random

Assumption 1: There are n identical random numbers $X_1, X_2, \dots, X_n$

Assumption 2: They are independent

Assumption 3: They have **finite** mean ($\mu = 0$) and variance ($\sigma^2 = 1$) values (*for simplicity*):

$$\mu_1 = \mu_2 = \dots = \mu_i = \dots = \mu_n = \mu = 0$$

$$\sigma_1^2 = \sigma_2^2 = \dots = \sigma_i^2 = \dots = \sigma_n^2 = \sigma^2 = 1$$

Claim: when n is large (infinity), their sum (or average) approaches a normal distribution:

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i \quad \longrightarrow \quad p(\sqrt{n}\bar{x}) = \frac{1}{\sqrt{2\pi}} e^{-(\sqrt{n}\bar{x})^2/2}$$

Central limit theorem

- Proof using moment generating functions

Moments definition:

Mean value for the k_{th} power of x

$$m_k = E [x^k] = \int_{-\infty}^{+\infty} x^k p(x) dx$$

Then by definition:

$$\mu = m_1 \quad \sigma^2 = m_2 - m_1^2$$

In our simple case:

$$\mu = m_1 = 0, \sigma^2 = m_2 = 1.$$

Moment generating function:

A convenient function that can be used to evaluate the value of every moment

$$M_x(t) = E [e^{tx}] = \int_{-\infty}^{+\infty} e^{tx} p(x) dx$$

Central limit theorem

- Proof using moment generating functions

$$M_x(t) = E[e^{tx}] = \int_{-\infty}^{+\infty} e^{tx} p(x) dx$$

How can it generate moment values? (Taylor expansion)

$$e^{tx} = 1 + tx + \frac{t^2 x^2}{2!} + \frac{t^3 x^3}{3!} + \dots + \frac{t^n x^n}{n!} + \dots$$

$$\begin{aligned} M_X(t) &= \int_{-\infty}^{+\infty} p(x) dx + \int_{-\infty}^{+\infty} txp(x) dx + \int_{-\infty}^{+\infty} \frac{t^2 x^2}{2!} p(x) dx + \dots + \int_{-\infty}^{+\infty} \frac{t^n x^n}{n!} p(x) dx + \dots \\ &= 1 + tE[X] + \frac{t^2 E[X^2]}{2!} + \dots + \frac{t^n E[X^n]}{n!} + \dots \\ &= 1 + tm_1 + \frac{t^2 m_2}{2!} + \dots + \frac{t^n m_n}{n!} + \dots \end{aligned}$$

To get the n th moment m_n :

Evaluate $M_x^n(t)$ at $t = 0$

$$M_x(t = 0) = m_0 = 1$$

$$M'_x(t = 0) = m_1$$

$$M''_x(t = 0) = m_2$$

.

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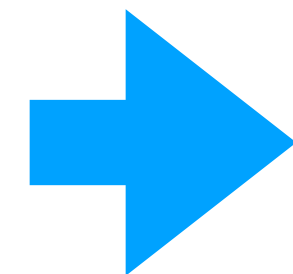
$$M_x^n(t = 0) = m_n$$

In the case of normal distribution

(Standard) normal distribution density function: $p(x) = \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2}$

Its moment generating function

$$\begin{aligned} M_x(t) &= \int_{-\infty}^{+\infty} e^{tx} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}x^2} dx \\ &= \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x^2 - 2tx + t^2)} e^{t^2/2} dx \\ &= e^{t^2/2} \int_{-\infty}^{+\infty} \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(x-t)^2} dx \\ &= \boxed{e^{t^2/2}} \end{aligned}$$



Total probability m_0

$$m_0 = e^0 = 1$$

Mean m_1

$$m_1 = te^{t^2/2} = 0$$

“Variance” m_2

$$m_2 = e^{t^2/2} + t^2 e^{t^2/2} = 1$$

Skewness m_3

$$m_3 = 3te^{t^2/2} + t^3 e^{t^2/2} = 0$$

Kurtosis m_4

$$m_4 = 3e^{t^2/2} + 6t^2 e^{t^2/2} + t^4 e^{t^2/2} = 3$$

Central limit theorem

- Proof using MGF (final step)

Mean of i.i.d random values

$$\bar{x} = \frac{1}{n} \sum_{i=1}^n x_i$$

Goal: prove that the moment generating function of the mean value is the same as normal distribution

MGF of normal distribution

$$M_{\sqrt{n}\bar{x}}(t) = e^{t^2/2}$$

Large $n \rightarrow +\infty$, then $\frac{t}{\sqrt{n}} \rightarrow 0$

$$M_{\sqrt{n}\bar{x}}(t) = E \left[e^{t\sqrt{n}\bar{x}} \right]$$

$$= E \left[e^{tx_1/\sqrt{n}} e^{tx_2/\sqrt{n}} \dots e^{tx_n/\sqrt{n}} \right]$$

Independent

$$= E \left[e^{tx_1/\sqrt{n}} \right] E \left[e^{tx_2/\sqrt{n}} \right] \dots E \left[e^{tx_n/\sqrt{n}} \right]$$

Identical

$$= \left(M_x \left(\frac{t}{\sqrt{n}} \right) \right)^n$$

$$= \left(1 + \frac{t^2}{2n} \right)^n$$

$$= e^{t^2/2}, \text{ when } n \rightarrow \infty$$

$$M_x \left(\frac{t}{\sqrt{n}} \right) \Big|_{\frac{t}{\sqrt{n}} \rightarrow 0} = M_x \left(\frac{t}{\sqrt{n}} \right) + \frac{M'_x \left(\frac{t}{\sqrt{n}} \right)}{1!} \left(\frac{t}{\sqrt{n}} \right) +$$

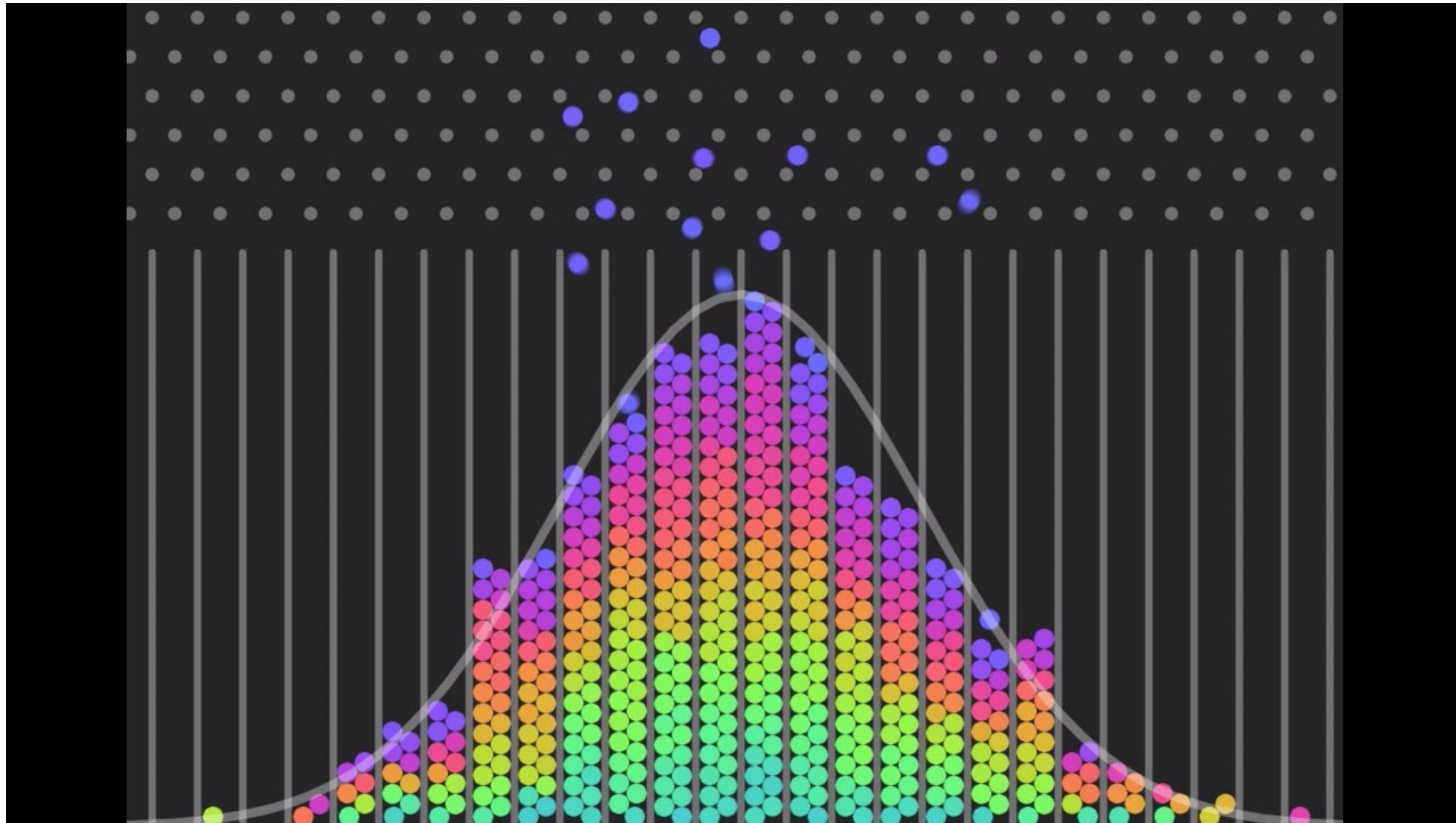
$$\frac{M''_x \left(\frac{t}{\sqrt{n}} \right)}{2!} \left(\frac{t}{\sqrt{n}} \right)^2 + O \left(\frac{t}{\sqrt{n}} \right)^3$$

$$= m_0 + m_1 \left(\frac{t}{\sqrt{n}} \right) + m_2 \frac{t^2}{2n} + O \left(\frac{t}{\sqrt{n}} \right)^3$$

$$= 1 + \frac{t^2}{2n}, \text{ since } m_1 = 0, m_2 = 1 \text{ and } n \rightarrow \infty$$

$$e^x = \lim_{n \rightarrow \infty} \left(1 + \frac{x}{n} \right)^n$$

Random walk



<https://www.youtube.com/watch?v=DxunFkBDdTA>

Exercise:

- Random walk leads to normal distribution

Probability density function

$$p(x) = \begin{cases} 0.5, & x = 1 \\ 0.5, & x = -1 \end{cases}$$

Moment generating function

$$\begin{aligned} M_x(t) &= \sum_x e^{tx} p(x) \\ &= \sum_{x=-1,1} e^{tx} 0.5 \\ &= \frac{e^{-t} + e^t}{2} \\ &= \cosh t \end{aligned}$$

Find the value of its moments

$$m_1 = \sinh 0 = 0$$

$$m_2 = \cosh 0 = 1$$

$$m_3 = \sinh 0 = 0$$

$$m_4 = \cosh 0 = 1$$

Practice (Python):

- 1. Simulate random walk of 5 steps, over 1000 simulations.
Fit the distribution against a normal distribution.**
- 2. Repeat 1 but for 25 steps random walk.**
- 3. Repeat 1 but for 125 steps random walk**